Labor Sheds

Technical Specification, Six Published Sizes and Quote Controls

Document	SP-LC-SHD-TS-R2
Revision	Rev 2 | 16 August 2026
Product boundary	Single-storey, one open sleeping hall under one continuous roof
Published sizes	20x10, 20x15, 20x20, 20x25, 20x30 and 20x40 ft
Commercial mode	Indicative ex-GST supply-only rate bands; no fixed per-size price table
Status	DRAFT FOR TECHNICAL SIGN-OFF

R2 resolves the earlier contradiction between a calculated size-price table and the approved quote-only product model. The six sizes are now treated as official LC-02 selections, not estimating reference bays. Bed counts remain subject to an approved bunk schedule, and no reviewer name or occupancy is invented in this draft.

Final structural member sizes, foundations, wind design, fire and life-safety provisions, electrical load schedule, exit arrangement, sanitation capacity and statutory approvals must be confirmed for the exact project before fabrication or use.

Published Size and Layout Schedule

All six selections are single-storey open halls. The arrangement notes describe the product configuration, not an approved worker capacity.

Code	Published size	Area	Approved arrangement	Door / circulation control
SZ-01	20x10 ft	200 sq ft	Single bunk row along one long wall	Clear side aisle; exits by approved GA
SZ-02	20x15 ft	300 sq ft	Two bunk rows against long walls	Continuous central aisle
SZ-03	20x20 ft	400 sq ft	Two rows; wider central aisle; end lockers	Distributed exits remain unobstructed
SZ-04	20x25 ft	500 sq ft	Two rows; central aisle	Second door at mid-point
SZ-05	20x30 ft	600 sq ft	Two rows; full-length central aisle	Doors at both ends
SZ-06	20x40 ft	800 sq ft	Two long rows; full-length central aisle	Three doors distributed along length

Mandatory product controls

- One continuous roof over one continuous internal volume; no room-by-room partitions.

- Continuous high-level louvre bands, protected ridge ventilator, gutters and downpipes.

- Exposed portal frames and purlins; no false ceiling in the baseline open hall.

- Sanitation, kitchen and wash facilities remain separate buildings unless a different product is engineered.

- RCC slab or designed hardstanding, safe external walkway, drainage and clear exit discharge routes.

Occupancy is deliberately not printed. A named technical reviewer must approve the bunk schedule, exit widths, travel distances and service layout before publication or fabrication.

Diagram 1: Open Dormitory Shed Plan

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Diagram 1: representative 20x30 ft open dormitory plan and six-size schedule

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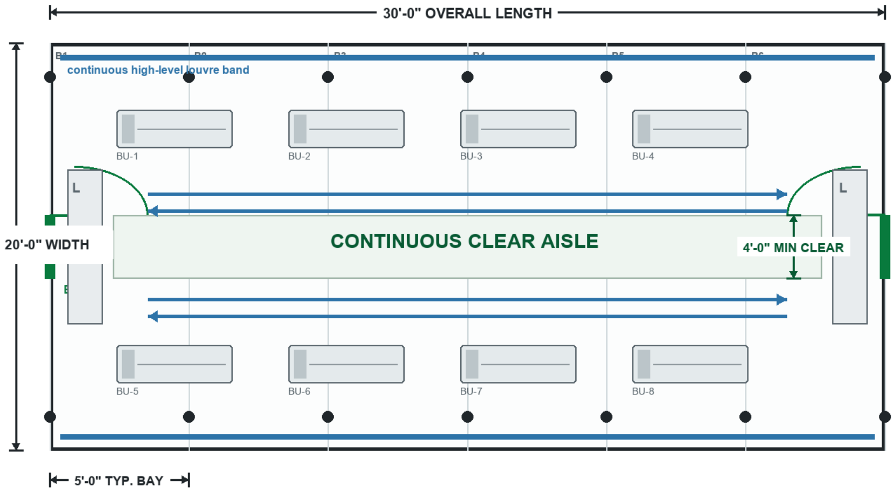
Open Dormitory Shed Plan

Representative 20x30 ft published size | single storey | one undivided sleeping hall



Illustrative fit-out shown: eight double-tier bunk modules. This is not an approved occupancy claim.

Door quantity, travel distance, aisle width, electrical points and bed count require the approved project GA and schedule.

Published size schedule

SZ-01	**20x10 ft**	200 sq ft	single bunk row
SZ-02	**20x15 ft**	300 sq ft	two rows + aisle
SZ-03	**20x20 ft**	400 sq ft	wider central aisle
SZ-04	**20x25 ft**	500 sq ft	mid-point exit
SZ-05	**20x30 ft**	600 sq ft	doors at both ends
SZ-06	**20x40 ft**	800 sq ft	three distributed exits

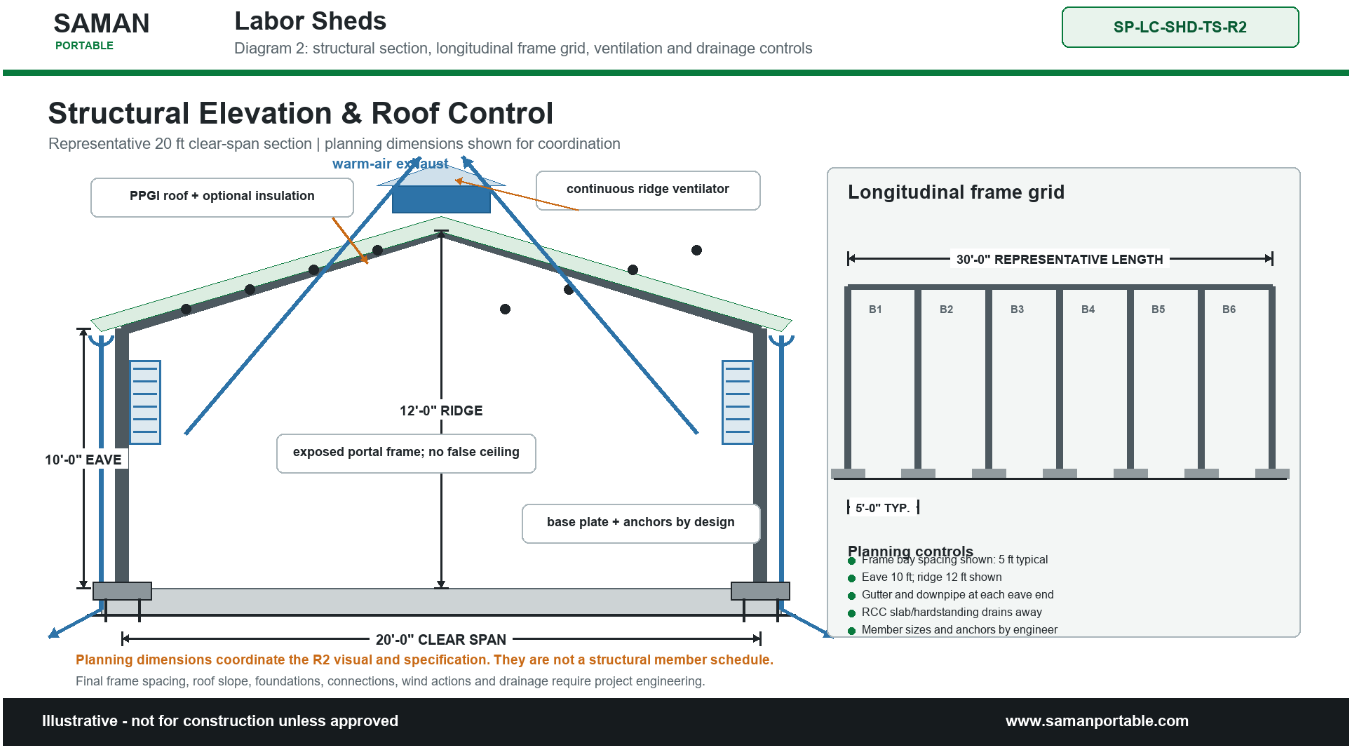
PLAN CONTROLS

- No internal partitions or false ceiling
- Sanitation remains in a separate block
- Keep every exit and aisle unobstructed
- Final bunk count requires approved schedule

Illustrative - not for construction unless approved www.samanportable.com

Representative plan: 20x30 ft, 5 ft frame bays and 4 ft minimum continuously clear aisle. Eight double-tier bunk modules are drawn only to explain circulation and coordination; they are not an approved bed count.

Diagram 2: Structural Elevation and Roof Control



Representative planning section: 20 ft clear span, 10 ft eave, 12 ft ridge and 5 ft typical longitudinal frame bay. These values coordinate the visual and specification; the project engineer must finalize members, connections, wind design, foundations and roof drainage.

Steel Structure

Component	Baseline specification	Coordination / design control
Bottom frame	IS 2062 or approved equivalent MS channel/RHS/formed sections; exact member sizes and support reactions from the signed structural drawing.	The base system transfers occupied, service, transport and local equipment loads to defined pads, plinths, slab or skid contact points. Anchor position, corrosion access and allowable bearing are frozen before fabrication; no nominal member is treated as universal across hutments, sheds, wet blocks and camp modules.
Bottom stiffeners	RHS/channel/angle stiffeners selected to the approved floor grid, partitions, bunks, wet services and lifting/relocation duty.	Stiffener spacing is coordinated with board edges, bunk posts, cubicle partitions, concentrated equipment and openings. Site cutting, member substitution and service penetrations require approval so the intended load path and corrosion protection are not lost.
Floor frame	RCC slab or designed hardstanding preferred; factory-floor cassette used only for smaller relocatable shed formats.	The slab supports dense bed rows and frequent cleaning without hidden board joints. Levelness, damp protection and any trench/services are coordinated before the shed frame is erected.
Top frame	MS RHS/channel or the documented proprietary upper rail sized for roof reactions, wall restraint and opening headers.	The upper frame closes the structural box, supports roof members and distributes wind and handling loads. Stair, canopy, stacked floor, tank, solar, exhaust or service-platform reactions are added only after project engineering.
Roof stiffeners	Light portal/truss frame with C/Z/RHS purlins supporting single- or double-slope PPGI roof; ridge/turbo ventilators and gutters as the airflow/drainage plan.	Span and uplift reactions are project engineered. Ventilator openings, roof laps and bracing remain clear of bed-zone obstructions and are maintainable from designated access points.
Corner posts / walls	Bolted or welded MS portal/column system with side rails and bracing; larger clear span than hutment modules.	Frame spacing is selected from shed width, cladding, wind and foundation reactions. G+1 or stacked images do not authorize multi-storey construction without separate structural/stair/egress design.
Lifting / handling	Marked lifting lugs, corner fittings, skid points or temporary transport brackets only where shown on the signed GA and lifting drawing.	The lift plan identifies finished weight, centre of gravity, sling geometry, spreader need and support points. One-piece lifting is not assumed for site-assembled sheds or panel camps, and forklift handling is allowed only when pockets and capacity are documented.
Welding & fabrication	IS 2062 / IS 1161 class structural steel or approved equivalent; weld sizes, bolt grades, hole tolerances and joint details as the fabrication drawings.	Fabrication checks cover fit-up, weld continuity, distortion, dimensions, bolt tightening, sharp edges and coating repair. IS 800 / project welding requirements are design references; certified WPS/PQR/NDT is supplied only when specified in the contract or QA plan.

Walls, Roof, Floor and Insulation

Component	Baseline specification	Coordination / design control
Exterior walls	PPGI/GI sheet with robust lower liner and continuous/high-level ventilation band, or insulated panel where climate/noise requires it.	The wall system prioritizes airflow along the sleeping hall. Insulated panels are an upgrade, not a blanket promise; low-level damage protection is coordinated with bed and circulation zones.
Roof	Profiled PPGI roof with defined overlap, ridge/eave details, gutters/downpipes and optional insulated underlay/panel.	Roof form supports continuous exhaust at ridge or approved ventilators. Rain ingress at ridge vents is controlled with baffles/flashings rather than sealing away the designed ventilation path.
Interior walls	Painted/galvanized sheet or durable low-wall liner; no decorative partition lining in the standard open dormitory.	Interior surfaces must tolerate repeated cleaning and impact near beds. Partitions are added only for supervisor/store or privacy functions shown on the GA.
Ceiling	Exposed insulated roof underside or simple radiant/liner treatment; ceiling only where comfort/acoustic/project scope justifies it.	Keeping the volume open supports stack ventilation and inspection. Any ceiling must preserve ridge airflow and provide safe access to fans/lights.
Floor base	RCC slab or specified industrial floor on prepared subgrade; no generic marine-ply platform for the standard large shed.	Floor flatness, damp control and perimeter drainage are site works. Local plinths below posts and partitions follow structural/foundation drawings.
Floor finish	Power-trowelled/sealed concrete, epoxy/PVC or other washable finish selected from project duration and cleaning regime.	Aisles, entrances and wash transition zones receive slip and wear consideration. Wet sanitation is kept in a separate block unless specifically integrated.
Wall insulation	PUF/sandwich-panel upgrade or insulated liner only when climate/occupancy requires it; GI tier may remain uninsulated with ventilation strategy.	The cost/comfort decision is visible in the quote. Insulation does not replace the need for ventilation at high occupant density.
Roof insulation	Insulated roof panel, reflective underlay or mineral/glass-wool class layer compatible with ridge ventilation and fire strategy.	Roof heat-gain measures are selected after climate and occupancy review. Field-added combustible lining is not accepted without approval.
Decorative / external finish	External fire/service access, paved or drained walkway, site lighting, stormwater path and connection to separate sanitation/drinking-water facilities.	The shed is treated as part of a compound. Bed-hall exits discharge to clear routes; sanitation, canteen and waste functions remain separated from the sleeping hall.
Fasteners & sealing	Corrosion-compatible fasteners, EPDM washers, butyl/PU sealants, closures, flashing, sleeves and isolation tapes matched to the selected wall/roof/floor build-up.	Joints and penetrations are detailed to drain outward, remain cleanable and allow maintenance. Sealant is not used to hide poor laps, unsupported joints or wet-area movement; field drilling is deburred, protected and resealed.

Doors, Windows, Electrical and Services

Component	Baseline specification	Coordination / design control
Main door / service door	Personnel doors/exits distributed along the hall based on approved occupancy, travel path and local requirements; robust MS/insulated doors as scheduled.	The layout prevents bunks from blocking exits and keeps door swings/landing areas clear. Exit count and width are not inferred from square footage in this workbook.
Windows / service opening	Continuous or repeated louvres/high windows on long elevations plus protected end openings, insect mesh and security features where required.	Openings create low-to-high and cross-flow ventilation over the sleeping hall while limiting driving rain. Window/louvre area is shown on the final elevation.
Grills / mosquito mesh	Removable mosquito mesh, louvres, restrictors, privacy screens or protected MS grilles only where required by room use and the approved opening schedule.	Accessories maintain ventilation, cleaning access and emergency escape. Sleeping areas, canteens and sanitation blocks use different privacy/security treatments; grille and mesh are not assumed on every opening.
Electrical wiring	Typical references: 1.5 sq.mm lighting, 2.5 sq.mm socket circuits and 4 sq.mm AC/high-load circuits; final cable sizing follows connected load, voltage drop and protection coordination.	FR/FRLS copper wiring is routed in approved conduit/trunking with protected metal penetrations and separation from water, waste, LPG/data and flexible module joints. Final cable size and derating are confirmed by the project electrical schedule.
Electrical protection	Main isolator, MCB/RCBO or RCCB protection, labelled neutral/earth bars, bonding and surge protection where required by the supply/risk assessment.	Steel frame and exposed conductive parts are bonded to the tested site earth. Wet-area circuits, generator/DG/UPS changeover, fault level and external supply interface follow the approved single-line diagram; this workbook does not certify the installation.
Electrical fittings	LED luminaires, fan points, 6A/16A sockets, exhaust points, emergency lighting and dedicated equipment outlets only in quantities shown on the approved room/electrical schedule.	Fixture density follows occupancy and function: sleeper, canteen, wet block, supervisor room and remote camp have different loads. Loose appliances, AC units, kitchen equipment, UPS/DG and data devices are excluded unless itemized.
Ventilation / AC	Cross-ventilation, ridge/stack exhaust, ceiling/wall fans and optional evaporative/mechanical assist selected from shed width, climate and occupant density.	The design avoids treating split AC as the default solution for a large open dormitory. Mechanical equipment is added only after heat/moisture and power review.
Plumbing / sanitary	Drinking-water and wash points nearby; soil/waste networks normally remain in a separate sanitation block rather than running through the sleeping hall.	Any internal basin or cooler point uses protected, maintainable pipe routes. The site plan establishes safe-water, drainage and sanitation capacity.
Layout / configuration	Parallel bed/bunk rows with central or dual aisles, distributed exits, cross ventilation and a separate common sanitation block.	Capacity follows an approved bunk and aisle schedule. The official R2 published sizes are 20x10, 20x15, 20x20, 20x25, 20x30 and 20x40 ft. No unapproved bed count is created by this workbook.
Painting / coating	Prepared carbon steel with compatible primer/top coats, or galvanized/pre-painted sheet with repaired cut edges; system upgraded for coastal, industrial or wet-service exposure.	The coating schedule records preparation, product, coat sequence, colour and repair method. Skid undersides, roof drains, cut edges, wet-area thresholds and inaccessible crevices receive inspection attention because these locations govern field durability.
Quality checks	Dimensional, member, weld/bolt, coating, panel, roof-drainage, opening, electrical and plumbing checks plus product-specific deployment or wet-test checks.	Handover evidence identifies the approved GA/BOM, material schedule, test records, photographs, punch items, lifting/service connection data and maintenance points. Fire, thermal, acoustic, structural or hygiene performance is stated only where supported by the contracted evidence.
Warranty	Warranty period, start date, remedy, relocation conditions, consumables and exclusions exactly as stated in the signed quotation and handover documents.	No universal warranty period is imposed by this workbook because attached and public product sources use different commercial packages. Site foundations, utilities, misuse, unauthorized alteration, poor cleaning, blocked drains, missed seal/coating maintenance and relocation damage remain excluded unless expressly covered.

Quote Bands, Included Scope and Exclusions

Shed floor area	R2 commercial route	Publication control
Below 100 sq ft	Use the PUF Porta Cabin product route	Not a Labor Shed; confirm current product-page price separately
100-1,000 sq ft	INR 600-750 per sq ft, ex-GST, supply only	All six published sizes fall in this band
Above 1,000 sq ft	INR 600-850 per sq ft, ex-GST, supply only	Larger clear spans quoted individually

Baseline supply scope

- Engineered light-steel shed enclosure to the approved drawing and bill of materials.
- PPGI/GI wall and roof cladding, continuous high-level louvre band, ridge ventilator, gutters and downpipes.
- Standard lighting, fans and ventilation provisions only where itemized in the electrical schedule.
- Factory and site quality checks within the signed quotation scope.

Normally excluded unless itemized

- Bunks, mattresses, lockers and loose furniture.
- Separate sanitation, water, drainage, kitchen and compound infrastructure.
- Foundations, slab/hardstanding, anchors, external utilities, fire systems and statutory approvals.
- Transport, unloading, craneage, positioning, installation and project-specific specialist testing.

The final written quotation must state the selected size, exact materials, openings, electrical quantities, exclusions, logistics, taxes, validity and warranty. No calculated size price from the withdrawn R1 model is valid for R2 publication.

Project Inputs, Quality Checks and Handover

Freeze before final drawing

- Project location, wind/exposure condition, service life and site levels.
- Selected size, quantity, bunk schedule, clear aisles, exits and external circulation.
- Door/louvre schedule, electrical load, lights, fans, emergency lighting and earthing interface.
- Foundation/slab, anchors, drainage, sanitation block, water and external utility interfaces.
- Transport route, unloading method, lifting restrictions and installation sequence.

Pre-dispatch and site quality checks

- Overall dimensions, frame identification, weld/bolt fit-up, coating and repaired cut edges.
- Roof laps, ridge ventilator, flashing, gutter falls, downpipe discharge and water-shedding details.
- Louvre continuity, door operation, sealing, electrical continuity, protection and frame bonding.
- As-built photographs, material schedule, approved deviations and punch-list closure.

Handover and maintenance

- Approved GA, bill of materials, opening/electrical schedules and maintenance points.
- Keep louvres, ridge vents, gutters and downpipes clear; repair leaks and coating damage promptly.
- Do not add openings, equipment supports or penetrations without design review.
- Warranty period, remedies, exclusions and relocation conditions follow the signed quotation only.

Document Control and R2 Resolution Record

R1 contradiction	R2 resolution
Smaller sizes called estimating reference bays	All six sizes are official LC-02 published selections
Calculated per-size prices shown	Removed from the customer-facing Labor Sheds model; quote bands only
Historical 20x30 to 30x50 wording	Superseded by 20x10 to 20x40 ft six-size schedule
Dimensionless diagrams	Representative plan and section now show feet, aisle, eave, ridge and bay dimensions
Unconfirmed occupancy risk	No bed count published; illustrative bunk modules clearly marked

Technical reviewer	PENDING - name, designation, signature and date
SEO / editorial reviewer	PENDING - name and date
Document status	DRAFT - do not publish or fabricate until technical sign-off
Supersedes	SP-LC-SHD-TS-R1 and its calculated per-size price table
Final authority	Signed quotation, approved project drawings, calculations and bill of materials

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